Place Value with Base-Ten Models
Answer Key

Grade 2 Math

Quick Answers

1. 135
2. 246
3. 308
4. hundreds = 5, tens = 7, ones = 3
5. 80
6. >
7. hundreds = 4, tens = 0, ones = 7
8. the number = 125, explanation = —

What is verified

12 of 13 answers machine-checked against SymPy, across 8 problems.
— marks an answer only you can judge — problem 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 2 Math
2.NBT.A – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 13 tagged checks.

Common wrong answers

3. *If they got 38: ignored the empty tens place and pushed the digits together*
5. *If they got 8: gave the digit itself instead of its value*

1. Write the number the model shows. (1 flat, 3 rods, 5 cubes)
Solution: The model shows 1 flat (1 hundred), 3 rods (3 tens), and 5 cubes (5 ones). That is $100 + 30 + 5$. $\boxed{135}$

2. A model shows 2 hundreds, 4 tens, and 6 ones. Write the number.
Solution: 2 hundreds is 200, 4 tens is 40, 6 ones is 6. Together: $200 + 40 + 6$. $\boxed{246}$

3. Write the number the model shows. (3 flats, no rods, 8 cubes)
Solution: The model has 3 flats (3 hundreds) and 8 cubes (8 ones) — and *no* rods at all, so the tens place holds a zero. The number is $300+0+8$. Watch for the zero: the answer is not 38. $\boxed{308}$

4. Fill in the blanks for the number 573.

Solution: Read each digit by its place: the 5 sits in the hundreds place, the 7 in the tens place, and the 3 in the ones place. So 573 has $\boxed{5}$ hundreds, $\boxed{7}$ tens, and $\boxed{3}$ ones.

5. What is the value of the tens digit in 482?

Solution: The tens digit of 482 is 8, and a digit in the tens place is worth ten times itself: $8 \times 10 = 80$. The digit is 8, but its *value* is $\boxed{80}$

6. Model A and Model B: write <, >, or =.

Solution: Model A shows 2 flats and 5 rods: $200 + 50 = 250$. Model B shows 2 flats, 3 rods, and 9 cubes: $200 + 30 + 9 = 239$. The hundreds match (2 and 2), so compare the tens: 5 tens beats 3 tens, and the 9 extra ones cannot make up a missing ten. $\boxed{250 > 239}$

7. Fill in the blanks for the number 407.

Solution: The 4 is in the hundreds place, the 0 is in the tens place (a placeholder — there are no loose tens), and the 7 is in the ones place. So $407 = \boxed{4}$ hundreds $+\boxed{0}$ tens $+\boxed{7}$ ones.

8. Challenge: a model shows 12 tens and 5 ones. What number is it? Explain the regrouping.

Solution: 12 tens is 120, and 5 ones is 5, so the model shows $120 + 5 = 125$. $\boxed{125}$

What a good explanation contains: ten of the twelve tens are traded for one hundred flat, which leaves 2 rods. After the trade the model shows 1 hundred, 2 tens, and 5 ones — the same amount, regrouped.